Directory Backup (s-DB.R)

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Description
This **R** script automates full directory backups from a source drive to an external disk. It includes built-in safety features such as dry-run simulation, automated pre-filter exclusions, native available disk space validation, long-path detection (>260 characters), and post-backup logging.

Instructions for Use:

1. Prerequisites:
Ensure you have **R** installed along with the required **fs** library. If the package is missing, install it in your **R** console:

```
install.packages("fs")
```

2. Configuration & Execution Steps:

1. Configure Script Variables: Set file paths and execution parameters.
Open the script and adjust the configuration section according to your setup:

- Set **source_dir** to the target drive/folder to back up (e.g., "E:/").
- Set **backup_base** to your destination folder path (e.g., "D:/STORAGE (E)").
- Set **report_dir** to store log files (e.g., "E:/BackupReports").

2. Run a Dry-Run Test: Verify safety before copying files.
Ensure **dry_run <- TRUE** is set in the configuration. Execute the script to simulate the backup process without writing files. Check the console output to confirm total files scanned and required space.

3. Execute Full Backup: Perform actual file transfer.

Set **dry_run <- FALSE** in the script configuration and run the entire script. Keep the **R** session active until execution completes.

4. Review Backup Logs: Check for truncated paths or skipped files.

Navigate to your configured report_dir to review the generated text logs:

- **backup_report_YYYY-MM-DD.txt**: High-level summary of total files processed and elapsed time.
- **file_paths_YYYY-MM-DD.txt**: Complete itemized list of scanned source files.
- **long_paths_YYYY-MM-DD.txt** *(if applicable)*: List of files that exceeded the **260**-character path limit and were skipped.